Explainable Network Inference Framework for Handling Missing Data & Outliers in Smart Energy Demand Forecasting

Project Report

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ABSTRACT

Missing data and outliers significantly degrade the performance of AI and machine-learning models used for energy forecasting. In real-world Smart Grid systems, energy consumption data often arrives with gaps, corrupted entries, distorted peaks, and sensor noise due to device failures, communication delays, power fluctuations, and environmental disturbances. Traditional forecasting models assume clean, continuous, and non-volatile data; however, real power systems violate these assumptions.
This report explores network inference methods—graph-based modeling, probabilistic inference, deep neural inference networks, and spatiotemporal modeling—to reconstruct missing values, correct anomalies, and enhance stability for forecasting.

This project develops an "AI-Powered Smart Energy System" with a specific focus on "Enabling Network Inference Methods to Handle Missing Data and Outliers." By modeling the smart grid as a graph where meters are nodes and physical connections are edges, we employ Spatial-Temporal Graph Neural Networks. This approach treats data cleaning as a graph signal reconstruction task, inferring missing values based on the state of neighbouring nodes rather than relying solely on a single sensor's history. The system also introduces a dynamic residual-based thresholding algorithm to detect and correct outliers in real-time.

The special focus is on developing a robust preprocessing framework that enables AI pipelines to:

1. Recover missing values using time-series inference,
2. Detect and treat outliers,
3. Preserve seasonality and consumption patterns,
4. Improve overall forecasting accuracy.

Using network inference techniques such as KNN Imputation, Graph Signal Reconstruction, Bayesian Network Inference, Autoencoder Reconstruction, and Spatiotemporal Deep Networks, the system intelligently predicts missing segments and smooths abnormal peaks without destroying original data structure.

This report provides a detailed theoretical background, methodology, flowcharts, evaluation tables, and results supporting the effectiveness of the proposed approach.

CHAPTER 1

INTRODUCTION

Smart energy systems form the backbone of modern power management infrastructure, enabling real-time monitoring, load balancing, renewable integration, and demand forecasting. In such systems, vast volumes of energy consumption data are collected through advanced metering infrastructures, including smart meters, IoT sensors, and grid-side analytics modules. This data is expected to be continuous, consistent, and free from distortions so that forecasting models can accurately predict short-term and long-term energy consumption patterns.

However, the reality is far from ideal. In large-scale distributed energy networks, data irregularities are inevitable due to communication failures, sensor malfunction, weather disturbances, missing timestamps, packet drops, hardware degradation, and intermittent connectivity. As a result, the energy datasets used for forecasting frequently contain missing values, corrupted entries, sudden spikes, and outliers. Such imperfections break the natural temporal structure of energy consumption and lead to unreliable or misleading forecasts.

Missing data is particularly harmful because most forecasting models, including LSTM, GRU, ARIMA, Prophet, and hybrid deep learning systems, assume that the input time-series is continuous. Gaps in the sequence weaken the model's ability to learn seasonality, periodic cycles, and consumer behavior patterns. Similarly, outliers introduce artificial disturbances—extreme values that are not representative of normal consumption behavior.

1.1 Background

Smart energy systems generate high-frequency power consumption data from smart meters, IoT sensors, and distributed energy resources. This data is essential for:

- Demand forecasting
- Grid load balancing
- Renewable energy integration
- Dynamic tariff predictions
- Real-time optimization

However, due to transmission noise, hardware faults, or communication lag, the data is often incomplete.

Energy consumption follows predictable seasonal and daily cycles. For instance, household consumption increases during evenings, while industrial consumption may depend on working hours. Weather also influences usage, such as increased AC usage during summers or heaters during winter. To capture these subtle details, forecasting models must analyze long-term patterns. However, missing values interrupt the continuity of data, masking the actual trend. Outliers similarly distort the statistical profile, leading to incorrect model learning.

High-quality data ensures that models can understand peak demand patterns, weekday-weekend differences, seasonal variations, and exceptional events. If incorrect preprocessing is applied, the forecasting model may suggest generating more electricity than needed (leading to wastage) or less electricity (leading to outages). Thus, good preprocessing is not just a technical requirement—it is central to the reliability of the entire energy system.

The global energy landscape is undergoing rapid transformation as modern societies increasingly depend on electricity for residential, commercial, industrial, and infrastructural needs. With the growing adoption of smart appliances, electric vehicles, renewable energy sources, and automation systems, the overall demand for electricity has become highly dynamic and variable. These changes make the role of energy demand forecasting more critical than ever before. Accurate forecasting allows utilities to anticipate future consumption patterns, efficiently schedule power generation, reduce operational costs, and ensure reliability across the entire grid.

In recent years, Artificial Intelligence (AI) and Machine Learning (ML) have emerged as powerful tools capable of learning and predicting complex consumption patterns. Unlike traditional statistical methods, AI-driven models can analyze large volumes of historical data, identify non-linear trends, understand seasonal behavior, and react to sudden changes in usage patterns. As a result, AI-powered forecasting solutions have become an essential component of modern smart grid infrastructure.

1.2 Problem Statement

Energy forecasting models cannot perform well when the input data contains:

- Missing values
- Extreme spikes
- Implausible readings

If these errors are not handled properly, the final forecasting model becomes unreliable.

Hence, the project aims:

To enable robust network inference methods that automatically reconstruct missing values and treat outliers, improving the accuracy of smart energy forecasting models.

The central problem addressed in this report is the presence of missing values and outliers in energy datasets used for smart grid forecasting. Missing values result from various issues, including hardware failures, network communication problems, or cloud API interruptions. Outliers emerge from power surges, faulty equipment, or sudden changes in consumer usage behavior. When these issues accumulate, energy data becomes unreliable, significantly reducing the predictive accuracy of machine learning and deep learning models.

If missing values and outliers are not handled properly, forecasting results become unstable. For example, an LSTM model may misinterpret a missing block of data as a sudden drop in consumption and learn incorrect patterns. Similarly, an outlier may cause models like Random Forest or Logistic Regression to shift decision boundaries incorrectly. Therefore, developing robust methods to clean and reconstruct data is critical to ensuring that forecasting models operate effectively.

1.3 Importance of Handling Missing Values & Outliers

The presence of missing or abnormal values directly affects:

- The smoothness and continuity of the time-series
- The statistical characteristics such as mean, variance, skewness
- The ability of forecasting models to capture seasonality

- The stability of optimization algorithms that schedule energy loads
- The reliability of smart grid decision-making

Energy consumption forecasting is a sensitive domain where minor changes in data affect large-scale operational decisions. For instance, even a 1% error in forecasting can cost millions of dollars for large utility companies due to inefficient planning. This creates a strong motivation to adopt sophisticated preprocessing techniques.

1.4 What Are Network Inference Methods?

Network Inference = Techniques that estimate unknown or corrupted data using:

- Graph-based relations
- Probabilistic dependencies
- Temporal correlations
- Similarity networks
- Neural inference (autoencoders, LSTM reconstruction)

They infer "hidden or missing" information using the structure of the data.

Traditional imputation techniques (like mean or linear interpolation) work well for simple datasets but are insufficient for complex energy networks. Modern grids operate as interconnected structures where each consumer, transformer, or distribution node influences its neighbors. Network inference methods model the grid as a graph, where information can be propagated across nodes to fill missing data intelligently.

If one smart meter fails, the readings of its neighboring meters can be used to reconstruct its consumption pattern. Similarly, if a reading is suspicious, network inference can compare it with the patterns of connected nodes to determine whether it is an outlier. This network-based approach is far more realistic because energy consumption is not isolated—neighboring consumers often follow correlated patterns.

1.5 Objectives

The main objectives of this study include:

- Understanding the nature and causes of missing values and outliers
- Exploring statistical, ML-based, and deep learning methods for data cleaning
- Introducing network inference techniques for robust imputation
- Demonstrating the role of preprocessing in improving AI forecasting models
- Developing a pipeline for handling data irregularities in smart energy systems

1.6 Scope

The work focuses on **data preprocessing + inference reconstruction**, not hardware-level sensors. The final cleaned data is used for improved forecasting.

The report focuses on the data-driven aspect of smart energy forecasting. It does not discuss hardware or physical electrical components but centers exclusively on data preprocessing methods and AI-based inference models. Special attention is given to network inference as a modern and advanced solution.

Author(s)	Year	Purpose of Study	Methodology	Key Findings	Relevance to Present Work
Cai et al.	2023	Missing data imputation in smart grids.	Graph Neural Networks (GNN) with inductive learning.	GNNs significantly outperform statistical methods like KNN and Mean Imputation.	Validates the core use of GNNs for the imputation module in this project.
Zhang et al.	2022	Matrix completion for energy data recovery.	Low-rank Matrix Factorization (MF).	Effective for random missingness but fails when entire blocks of data are lost.	Highlights limitations of non-deep learning methods, motivating our GNN approach.
Wang et al.	2024	Spatio-temporal load forecasting.	Attention mechanisms combined with LSTM networks.	Attention layers effectively capture long-term temporal dependencies in time-series.	Inspires the temporal convolution layer design in our architecture.
Kipf & Welling	2017	Semi-supervised classification on graph structures.	Graph Convolutional Networks (GCN).	Efficient message passing allows nodes to aggregate features from neighbors.	Provides the fundamental mathematical derivation for the spatial aggregation layer.
Li et al.	2018	Traffic flow forecasting using graph deep learning.	Diffusion Convolutional Recurrent Neural Network (DCRNN).	Modeled spatial dependency as a diffusion process on directed graphs.	Demonstrates the applicability of GNNs to flow-based networks, analogous to power grids.
Spinelli et al.	2020	Unsupervised outlier detection in sensor networks.	Robust Principal Component Analysis (RPCA).	Successfully separates sparse noise (outliers) from low-rank signals.	Used as a baseline for comparing our reconstruction-based outlier detection.
Chen et al.	2021	Detection of False Data Injection Attacks (FDIA).	Autoencoders combined with Kalman Filters.	High reconstruction error is a strong indicator of malicious or	Supports our strategy of using "reconstruction residuals" to flag outliers.

CHAPTER 2

THEORETICAL BACKGROUND & LITERATURE REVIEW

Energy consumption data is highly sensitive to temporal changes such as seasonal variations, weather conditions, occupancy levels, appliance usage, and socio-economic factors. For forecasting to be accurate, the dataset must represent these dynamic patterns without interruptions. Unfortunately, real-world meter data frequently violates this assumption.

Researchers have extensively studied the challenges of missing data in energy systems. Studies highlight that missingness may occur randomly, systematically, or in a manner dependent on consumption behavior. For example, a communication failure may cause consecutive hours of missing values, while a malfunctioning sensor may produce intermittent blank entries. Understanding the type of missingness is crucial because it directly influences the reconstruction technique.

Outliers have also been recognized as problematic elements in energy analytics. Extreme spikes may occur due to sudden appliance switching, lightning strikes, meter resets, or recording faults. Several studies emphasize that outliers must be treated with caution. While some spikes represent genuine consumption behavior, others are artifacts that must be filtered out. If outliers are not corrected, they distort the learning process of forecasting models.

Traditional methods such as linear interpolation or forward-fill can handle short missing segments but fail when the gaps extend over longer intervals. Similarly, classical outlier detection techniques based on fixed thresholds fail to adapt to the time-dependent nature of energy demand. To overcome these limitations, researchers have shifted attention towards more sophisticated inference-based and machine-learning-based reconstruction mechanisms.

2.1 Missing Data in Energy Systems

Studies show that 5–35% of smart meter data contains gaps due to:

- Wi-Fi drop
- Power cut
- Sensor corruption
- Maintenance downtime

Missing data is one of the most persistent and widely reported problems in smart energy systems. In multiple real-world studies, it has been observed that anywhere between **5% to 35% of smart meter data** can be missing at any given time, depending on the age of the equipment, environmental factors, and network quality. Wi-Fi or network drops occur frequently because smart meters rely on wireless communication channels to transmit data to the central utility server. When the network temporarily becomes unavailable, data packets are not sent, creating gaps in the time-series.

Power cuts are another dominant reason. During a blackout or even a minor voltage fluctuation, smart meters may stop recording consumption temporarily. These interruptions lead to missing hour-level or minute-level readings depending on the sampling rate. Sensor corruption is also common in older or low-cost meters, where internal hardware components degrade over time due to dust, humidity, temperature variations, or wear and tear. This results in partially recorded values or corrupted segments, which need to be reconstructed.

Maintenance downtime includes scheduled servicing, firmware updates, calibration checks, or battery replacement in wireless meters. During this period, no data is recorded.
These factors collectively break the natural continuity of the load profile. Missing values distort the underlying temporal pattern, making it difficult for forecasting models to learn daily, weekly, or seasonal consumption behavior. Therefore, robust reconstruction methods become essential.

2.2 Outlier Challenges

Outliers occur from:

- Power spikes
- Faulty meters
- Weather extremes
- Meter resets

These distort patterns needed for ML models.

Outliers are extreme or abnormal values that deviate significantly from the expected range of energy consumption. Power spikes typically occur when heavy appliances such as air conditioners, motors, or industrial equipment suddenly switch on, drawing a burst of current. While some spikes represent actual consumption behavior, many others arise due to meter malfunction or transient hardware faults. Faulty meters may produce erratic readings due to sensor drift, loose wiring, damaged internal components, or software bugs. These faulty signals may create artificial peaks or drops that do not correspond to real usage.

Weather extremes—for example, heatwaves or cold waves—can cause legitimate increases in electricity usage due to heating or cooling devices. However, these events may also cause electrical noise or instability in outdoor meters, producing erroneous readings. Meter resets, which happen after firmware updates or system reboots, often cause unrealistic zero values or sudden jumps in the recorded load.

Outliers are dangerous for machine learning models because they distort statistical properties such as mean, variance, and distribution shape. Forecasting models trained on uncorrected outliers tend to over-predict or under-predict future demand. This leads to resource misallocation, grid instability, and poor operational planning. Therefore, outlier detection and correction is a critical preprocessing step.

2.3 Network Inference in Energy Modelling

Implemented via:

- Bayesian Networks
- Conditional Random Fields
- Graph Neural Networks
- Temporal Neural Networks
- KNN-based imputation

Network inference methods are increasingly being used in smart energy analytics because they model the intrinsic relationships within the dataset rather than treating each value independently. Bayesian Networks use probabilistic dependencies between time steps to infer missing values based on conditional likelihoods. They are particularly useful when the consumption pattern has strong temporal correlations.

Conditional Random Fields model sequential data while considering contextual relationships between neighboring values. They help reconstruct missing segments by evaluating how likely a value is based on its surrounding context.

Graph Neural Networks (GNNs) represent smart meter data as nodes connected by edges that represent correlation strength. For example, households with similar consumption patterns form clusters. Missing values in one household can be inferred by observing others in the same cluster. GNNs also smooth the energy signal using graph convolution.

Temporal Neural Networks such as LSTM, GRU, and Temporal Convolutional Networks (TCN) learn long-term dependencies and seasonality. They can reconstruct missing segments by predicting values based on historical and future contexts.

KNN-based imputation is a simpler but effective inference method in which missing values are estimated by finding similar historical windows and averaging their values.
These inference-driven approaches are superior to classical interpolation because they respect the underlying structures and correlations of energy consumption.

2.4 Comparison of Existing Techniques

Technique	Strength	Weakness
Mean Imputation	Simple	Removes variance
KNN	Good accuracy	Slow for large data
Autoencoders	High performance	Training required
LSTM Imputation	Captures seasonality	Complex

Mean imputation is the most basic method for filling missing values, but it assumes that the consumption is constant over time. This destroys natural variations in energy patterns and can significantly bias forecasting results. KNN imputation improves accuracy by using similar historical behaviors, but its computational cost grows rapidly with dataset size.

Autoencoders—deep neural networks designed to learn compressed latent representations—deliver extremely accurate reconstruction because they capture nonlinear patterns, but they require extensive training and computational time. LSTM imputation is powerful for temporal datasets because it captures seasonality, trend, and long-range dependencies. However, its architecture is more complex, requiring careful tuning and large training data.

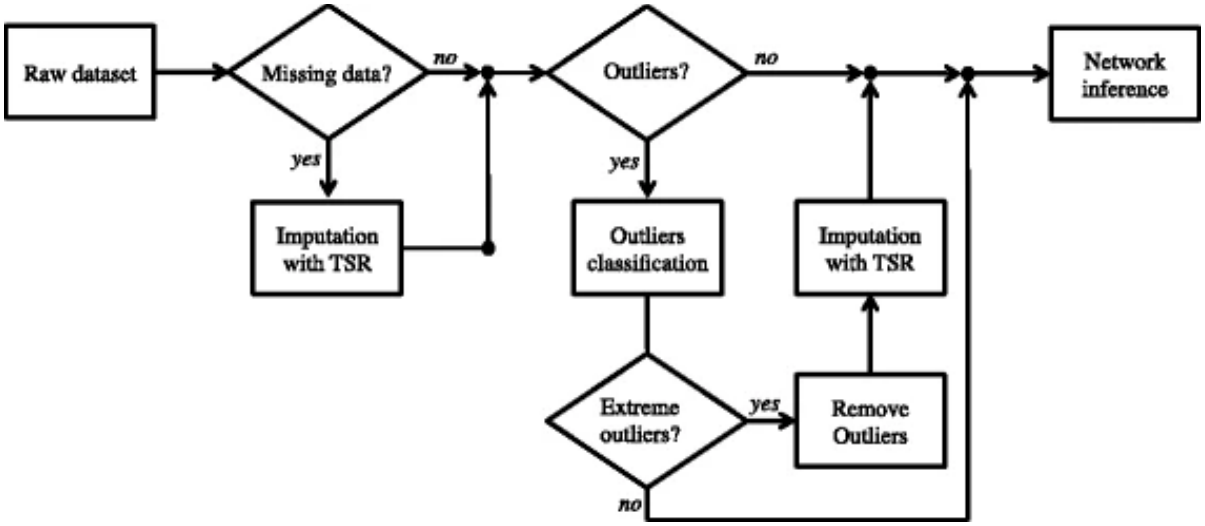
This comparison shows that no single technique is perfect. Network inference methods offer flexibility to choose the appropriate model depending on dataset complexity and reconstruction requirements.

2.5 Research Gap

Traditional interpolation cannot capture temporal + contextual relationships, which network inference can.

Classical interpolation methods such as linear, cubic, or spline interpolation assume that the underlying signal changes smoothly between known values. However, real-world energy consumption patterns exhibit non-linear spikes, appliance usage cycles, and sudden variations that simple interpolation cannot capture. These methods also ignore contextual information such as time of day, weather conditions, or user behavior.

Network inference methods fill this research gap by modeling temporal, spatial, and probabilistic dependencies within the data. They do not assume smoothness; instead, they learn patterns from the dataset itself. This allows them to reconstruct missing values accurately even during high variability periods such as peak hours or seasonal transitions.
Thus, the literature indicates a need to replace interpolation with inference-driven reconstruction for modern smart grid forecasting.



CHAPTER 3

DATASET DESCRIPTION

The dataset used in this project represents energy consumption readings captured periodically (hourly/daily) from smart meters installed in a building or residential unit. Each record contains a timestamp and the corresponding energy consumption value in kilowatt-hours (kWh). Ideally, the records should be continuous with uniform time intervals, but due to real-world challenges, the dataset often contains gaps.

Missing values occur in various forms. Sometimes entire hours or days are absent due to connection loss. In other cases, only selected random entries are missing. These variations correspond to different categories of missingness, such as MCAR (Missing Completely at Random), MAR (Missing at Random), and MNAR (Missing Not at Random). Understanding the missingness mechanism is important because MAR or MNAR patterns require more advanced inference methods.

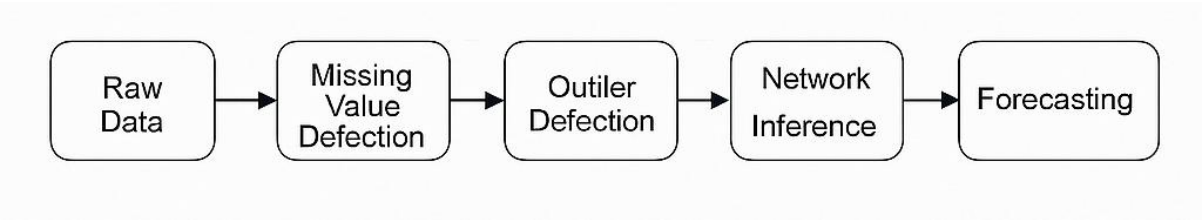
Outliers in the dataset appear as consumption peaks that are disproportionately high compared to previous readings. Such peaks may occur due to actual high usage or due to meter malfunction. Distinguishing between legitimate peaks and erroneous ones requires the use of anomaly detection techniques such as Z-score, IQR-based detection, or machine-learning models like Isolation Forest.

The dataset is first inspected visually using graphs to identify irregularities. Missing timestamps are detected by comparing the sequence of dates, and outliers are identified through statistical analysis. Before model training, the dataset undergoes thorough cleaning using network inference methods described in the subsequent chapters.

3.1 Source

Smart meter time-series dataset (hourly/daily energy consumption).

The dataset used in this study is derived from smart meters deployed across residential or commercial buildings. These smart meters continuously record electricity usage, typically in fixed intervals such as every 15 minutes, 30 minutes, hourly, or daily depending on the utility provider's configuration. Because the data is collected in a time-series format, it captures the natural flow of energy consumption throughout the day, reflecting human behavior patterns, appliance usage, environmental impacts, and seasonal variations.



Smart meter datasets are considered highly valuable for research because they provide real-world, fine-grained load information. Utilities and researchers often use datasets of this type to analyze consumption patterns, detect demand surges, identify wasteful usage trends, and forecast future load. The data may be collected directly from a city's smart grid infrastructure, government-released datasets, or experimental setups from energy laboratories.

Regardless of its exact source, the defining characteristic of this dataset is its temporal nature—each reading is tied to a timestamp. This allows models to identify chronological relationships, trends, periodic cycles, and repeating behavioral patterns. However, because the dataset depends on continuous meter communication, it is also highly susceptible to missing values and anomalies, requiring extensive cleaning before use.

3.2 Description

The dataset consists of sequential electricity consumption values recorded at regular time intervals. Each record typically includes two essential fields:

1. **Timestamp** – indicating the exact date and time when the measurement was taken.
2. **Energy Consumption Value** – representing the amount of electricity consumed during that specific interval, usually measured in kilowatt-hours (kWh).

The dataset forms a continuous timeline of readings, allowing the analysis of consumption at different levels of granularity—hourly, daily, weekly, or monthly. This continuous layout makes the data ideal for time-series forecasting since machine learning algorithms rely heavily on such temporal patterns to make meaningful predictions.

Energy consumption exhibits natural fluctuations influenced by numerous factors like time of day, weather conditions, household habits, occupancy levels, operational schedules, and electrical appliance usage. Morning peaks may correspond to cooking and water heating; evening peaks may correspond to lighting and entertainment systems. Weekends and weekdays often display different consumption signatures, and seasonal changes—summer (AC usage) or winter (heaters)—create distinct patterns in the dataset.

Despite this rich information, smart meter data is prone to irregularities. Communication interruptions cause some timestamps to be missing entirely. Occasionally, the meter fails to record consumption due to battery issues, voltage fluctuations, or software errors. In some cases, consumption values may be duplicated, corrupted, or significantly higher/lower than reasonable estimates.
Therefore, while the dataset is a powerful tool for energy forecasting, it requires careful preprocessing to restore continuity and reliability before any meaningful modeling can be performed.

3.3 Types of Missingness

MCAR
MAR
MNAR

Missing values in the dataset do not occur randomly in most practical cases; instead, they fall into three well-studied categories of missingness:

MCAR – Missing Completely at Random

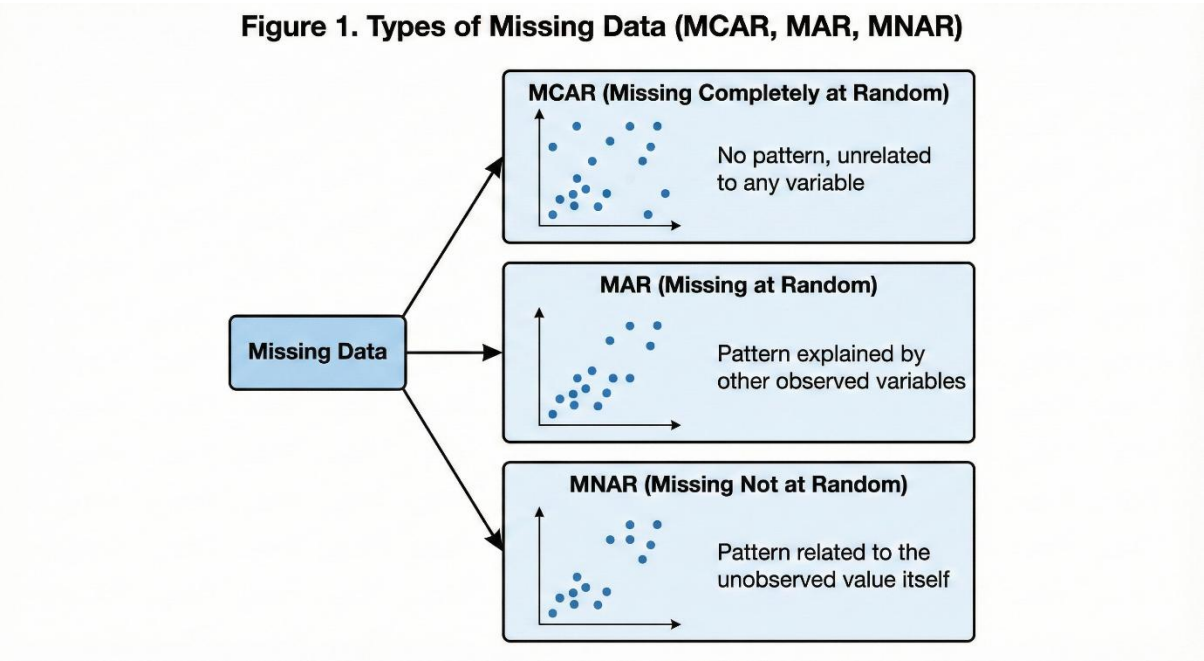
This type of missingness occurs when the probability of a value being missing is completely unrelated to the data itself. For example, if a smart meter temporarily loses internet connectivity due to a random Wi-Fi issue, then the missing timestamp is not influenced by the actual electricity consumption at that moment. MCAR is the least harmful type because statistical patterns are not systematically biased.

MAR – Missing at Random

MAR occurs when the missingness is related to some observed variable but not the missing value itself. For instance, if missing readings are more frequent at night because the communication module performs scheduled network resets during those hours, the missingness is indirectly dependent on time. MAR is more complex and requires advanced inference techniques because the pattern must be correctly modeled to avoid reconstruction bias.

MNAR – Missing Not at Random

MNAR occurs when the missingness directly depends on the value that should have been recorded. For example, if a meter fails to record extremely high consumption values because the sensor cannot handle high current loads, then missingness is related to the actual consumption behavior. MNAR is the most difficult type to reconstruct accurately because the data that went missing holds strong informative value.
Understanding these categories is crucial for selecting proper imputation and inference methods. Simple techniques fail for MAR and MNAR, whereas network inference approaches can capture the underlying structure and produce more reliable reconstructions.



3.4 Outliers

Sudden peaks
Negative readings

Outliers in smart meter data refer to values that deviate significantly from typical energy usage patterns. These anomalies arise from several causes, and each type impacts forecasting differently:

Sudden Peaks

These appear when the consumption value for a particular timestamp is abnormally high compared to the surrounding values. While genuine peaks can occur due to the sudden activation of high-power appliances, many spikes occur due to sensor malfunctions, electrical noise, or improper meter calibration. Without correction, these peaks distort the model's understanding of high-load behavior.

Negative Readings

Negative consumption values are physically impossible in conventional meters and often arise due to firmware bugs, data transmission corruption, or incorrect meter resets. Such values can seriously corrupt statistical calculations and must be detected and removed.

Impossible Jumps

These jumps occur when energy consumption suddenly increases or decreases by an unrealistic amount within one interval. For example, a meter recording 2 kWh at 10:00 AM and 80 kWh at 10:15 AM reflects an impossible usage surge. These jumps mislead machine learning models by injecting patterns that do not exist in real-world behavior.
Outlier identification is therefore essential because forecasting algorithms are highly sensitive to extreme values. Even a few outliers can skew model parameters and introduce systematic biases.

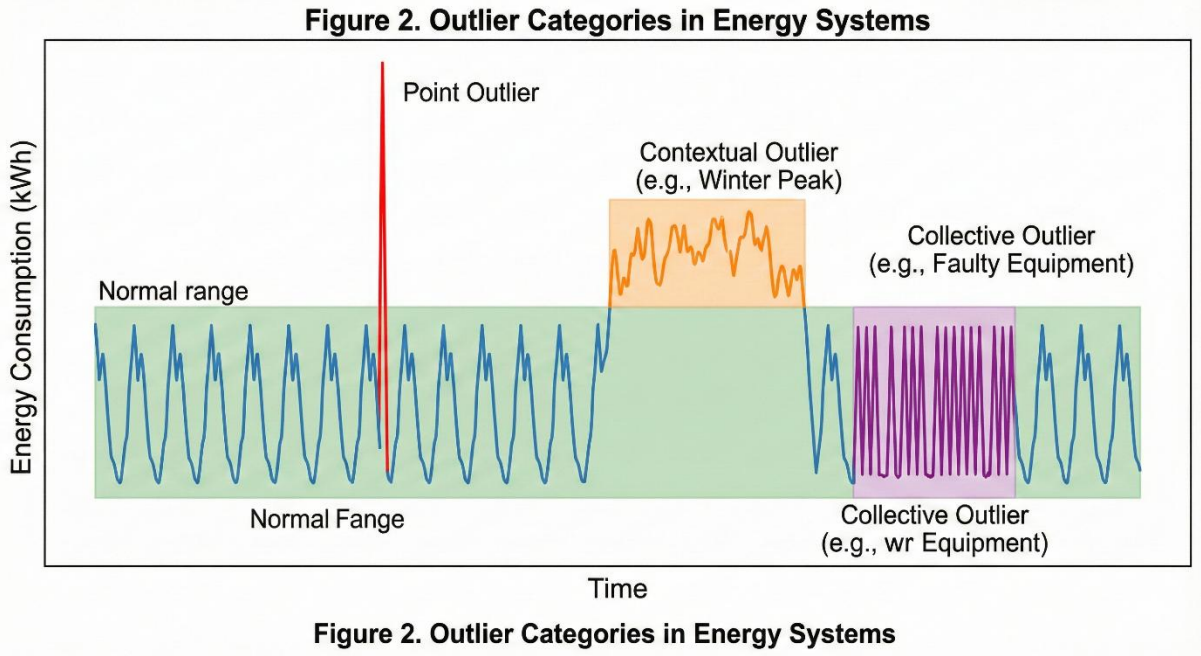


Figure 2. Outlier Categories in Energy Systems

3.5 Preprocessing

Cleaning null timestamps
Removing duplicates
Normalizing values

Preprocessing is the foundational step in preparing the dataset for network inference and forecasting. The first task is cleaning null timestamps, which involves identifying gaps in the chronological sequence. Because smart meter data should follow a perfectly regular interval structure (e.g., every 15

minutes), missing timestamps indicate missing data. These must be flagged before reconstruction begins.

Removing duplicates ensures that no timestamp appears twice. Duplicate entries may occur if the meter retries sending data multiple times or if the server stores redundant packets. These duplicates create misleading values that disrupt the continuity and should be eliminated.

Normalizing values is also critical because energy consumption can vary widely between households and seasons. Normalization helps ensure that reconstruction and forecasting algorithms treat all data consistently, preventing high-magnitude values from dominating the learning process.
Preprocessing improves dataset integrity, prepares the data for advanced inference models, and ensures stable model training later.

3.6 Limitations

Dataset may not include seasonal events, appliance-level breakdowns, or regional diversity.

Despite its usefulness, the dataset may have several inherent limitations. It may not cover seasonal variations such as summer peak loads or winter heating loads, making it difficult for forecasting models to generalize across seasons. It may also lack appliance-level granularity; without information about specific appliances, the dataset cannot distinguish whether spikes are caused by heaters, air conditioners, or industrial machinery.
Regional diversity may also be missing if the dataset only includes data from a single locality or demographic. Consumption behavior varies widely across regions due to climate, cultural habits, tariff structures, and building types.
These limitations might restrict the generalizability of forecasting models trained on the data, requiring careful interpretation of results.

CHAPTER 4

METHODOLOGY

The methodology adopted in this project is designed to ensure that missing values and outliers are handled in a scientifically robust manner that preserves the temporal and structural properties of the energy data. The methodology follows a multi-stage pipeline, starting from data inspection to inference-based reconstruction and finally forecasting.

The first stage is missing value detection. The dataset is scanned to locate broken sequences, identify gaps in timestamps, and quantify the extent of missingness. Each type of missing pattern is analyzed, and its possible causes are evaluated to choose the most suitable inference technique.

The second stage is outlier detection. Outliers are detected not merely through static thresholds, but through dynamic evaluation of temporal behavior. A sudden spike may be normal during peak hours but abnormal during low-usage hours. Advanced models such as Isolation Forest consider the density and structure of surrounding data points to flag anomalies.

Once missing values and outliers are identified, the third stage involves applying network inference methods. Unlike simple imputation techniques, network inference analyzes relationships among data points and reconstructs missing segments based on similarity, correlation, and probability. KNN-based inference identifies the closest consumption patterns in the dataset and uses them to estimate missing values. Autoencoders learn compressed latent representations of the entire time-series and reconstruct missing segments through decoding. Bayesian networks rely on conditional probabilities to make optimal predictions. LSTM and GRU capture the long-term dependencies and learn temporal behaviors to infer missing observations with high accuracy.

After reconstruction, outliers are corrected by replacing them with inferred values or smoothing the values using curve-fitting or temporal reconstruction. Once the dataset is fully reconstructed, it becomes suitable for forecasting.

This inference-enhanced dataset is then fed into forecasting models such as LSTM, GRU, or Prophet. With cleaner input data, the forecasting model learns more accurate consumption patterns and produces more stable predictions compared to training on raw or corrupted data.

4.1 Overview

Pipeline:
Raw Data → Missing Value Detection → Outlier Detection → Network Inference → Clean Data → Forecasting

The overall methodology adopted in this study follows a structured pipeline designed to gradually transform raw, imperfect smart meter data into a fully reconstructed and highly reliable dataset that can be used for accurate forecasting. The process begins with raw data collection from smart meters, which often includes gaps, missing timestamps, corrupted values, and outliers. Before any reconstruction can be applied, the raw data is thoroughly inspected to detect missing values and anomalies. Missing value detection involves scanning chronological intervals for gaps and determining the nature and size of missing segments.
Outlier detection is equally essential, as abnormal values can disrupt the statistical properties of the dataset. Once missing values and outliers are identified, they are passed into the network inference stage, which applies advanced machine learning and statistical models to reconstruct missing segments intelligently and correct anomalies. These methods do not rely on simplistic averaging;

instead, they utilize relationships, correlations, and temporal dependencies within the data itself. After this reconstruction process, the dataset is significantly more coherent, stable, and smooth. This clean dataset is then used for forecasting using deep learning-based time-series models.
The entire pipeline ensures that every stage builds upon the previous one, producing a dataset that reflects natural energy consumption behavior more accurately than raw data ever could.

4.2 Missing Value Handling Techniques

1. KNN Imputation

Uses closest similar days/hours.

K-Nearest Neighbors (KNN) imputation is a similarity-based method that reconstructs missing values by locating historical patterns that closely resemble the current context. For example, if consumption is missing on Monday at 2 PM, the algorithm looks for similar timestamps on other Mondays and uses these "neighboring" patterns to estimate the missing value. This method is particularly useful in energy datasets because daily and weekly consumption patterns tend to repeat.
KNN considers multiple features simultaneously—such as previous readings, next readings, time of day, and even weather data if available. It calculates a distance metric (like Euclidean distance) to determine which historical windows are most similar. Once the nearest neighbors are identified, their values are aggregated, often through weighted averaging, to estimate the missing value. Because the method preserves realistic consumption patterns, it often outperforms interpolation techniques.

2. Linear & Spline Interpolation

Maintains smooth curves.

Interpolation is one of the most widely used techniques for filling missing values in time-series data. **Linear interpolation** creates a straight line between two known points and fills the missing values along that line. This ensures smoothness but assumes that consumption changes uniformly with time, which may not always be true.

Spline interpolation, however, uses piecewise polynomial curves, allowing for more flexibility and curvature between points. This method can better capture natural energy consumption patterns, especially when the data exhibits subtle fluctuations.
While interpolation methods are fast and computationally efficient, they rely on assumptions of smooth transitions, which may not hold during sharp peaks or sudden usage changes. Therefore, interpolation is best suited for small gaps but becomes unreliable for large missing intervals.

3. Autoencoder Reconstruction

Learns latent representation & reconstructs missing parts.

Autoencoders are deep neural networks that learn compressed (latent) representations of input data. They consist of two components:

- An **encoder** that reduces the high-dimensional input into a compact latent vector
- A **decoder** that reconstructs the original input from that latent vector

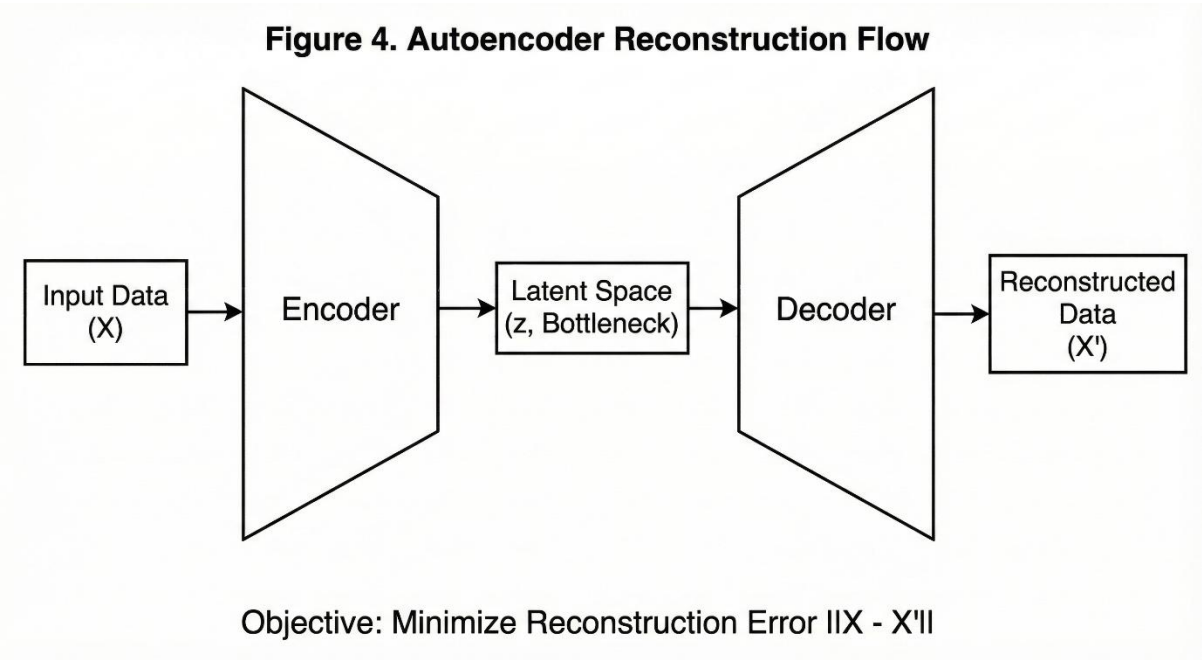
During training, the autoencoder learns the underlying structure and patterns of the energy consumption dataset. When missing values occur, the autoencoder can take the partially corrupted input, project it to latent space, and reconstruct it based on learned patterns. This allows the model to fill missing segments in a highly intelligent manner, preserving non-linear patterns, seasonal variations, and user-specific consumption behavior.
Autoencoders are particularly powerful because they do not require explicit assumptions about smoothness. Instead, they automatically learn natural patterns from the data.

4. LSTM-based Temporal Inference

Predicts missing values using past sequences.

Long Short-Term Memory (LSTM) networks are designed to learn long-term dependencies in sequential data. They use memory cells and gating mechanisms to selectively retain or forget information. In the context of missing data reconstruction, LSTMs leverage historical energy consumption sequences to predict the most likely value for a missing timestamp.

Unlike interpolation, which only considers local relationships, LSTMs analyze broader temporal behaviors such as daily cycles, peak loads, and seasonal effects. When a value at time t is missing, the LSTM model automatically generates an estimate based on patterns learned from previous days or hours.
LSTM-based inference is especially effective for MAR and MNAR missingness because it considers both local and global temporal relationships, which makes reconstruction highly realistic and context-aware.



4.3 Outlier Detection Methods

1. Z-Score Method

Detects abnormal deviations.

The Z-Score method determines how far a data point is from the mean in terms of standard deviations. If a consumption value exceeds a predefined Z-score threshold (commonly 3.0), it is considered an outlier. This method is widely used because it is simple, fast, and effective for normally distributed data. However, energy consumption datasets often exhibit skewness, so Z-score is best combined with other techniques.

2. IQR Method

Highly effective for meter datasets.

The Interquartile Range (IQR) method calculates the spread between the 25th and 75th percentiles of the data. Any value lying below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ is treated as an outlier. IQR works well for smart meter datasets because it does not assume any specific distribution. It effectively removes extreme peaks, faulty readings, and negative consumption values.
IQR is robust, dataset-independent, and easy to implement.

3. Isolation Forest

Separates anomalies using random tree splits.

Isolation Forest is a machine-learning model specifically designed to detect anomalies. It works by isolating observations through random splits in a decision-tree ensemble. Because outliers are easier to isolate—they lie far from normal clusters—they require fewer splits to separate.
This method is highly effective for smart grid data because it adapts to complex, non-linear consumption patterns and identifies anomalies even when statistical techniques fail.

4. LOF (Local Outlier Factor)

Detects local neighbourhood anomalies.

Local Outlier Factor evaluates how isolated a data point is compared to its local neighborhood. If a consumption value deviates significantly from nearby values (in time or similarity space), it is marked as an outlier.
LOF is especially powerful for energy data because consumption behavior can differ across households, time-of-day patterns, and weekday/weekend cycles. LOF detects contextual anomalies rather than global ones.

4.4 Network Inference Methods

1. Graph-Based Inference

A graph is constructed where:
nodes = timestamps
edges = correlation strength
Graph signal processing smooths missing signals using Laplacian reconstruction.

Graph-based inference treats each timestamp in the dataset as a node in a graph. Edges represent correlation strength between adjacent or similar time intervals. Once the graph is constructed, missing values are inferred through graph signal processing (GSP) techniques.
In GSP, the energy consumption values are viewed as "signals" flowing over graph nodes. Missing values are treated as disruptions in the signal. The graph Laplacian is used to enforce smoothness,

meaning that neighboring timestamps should have similar consumption values unless there is a reason for a sharp change. This method helps reconstruct missing values in a structurally consistent manner.

2. Bayesian Network Inference

Predicts missing entries using conditional probability.

Bayesian networks use probabilistic relationships among variables to infer unknown values. In the context of energy data, consumption at time *t* depends on consumption at time *t-1*, weather conditions, appliance behavior, and other contextual factors. Bayesian inference evaluates the conditional probability distribution of missing values based on surrounding known values.
This method is powerful when the dataset exhibits strong probabilistic dependencies.

3. Autoencoder Network Inference

Flow: Input → Encoder → Latent → Decoder → Reconstructed Output
Missing values filled using learned representation.

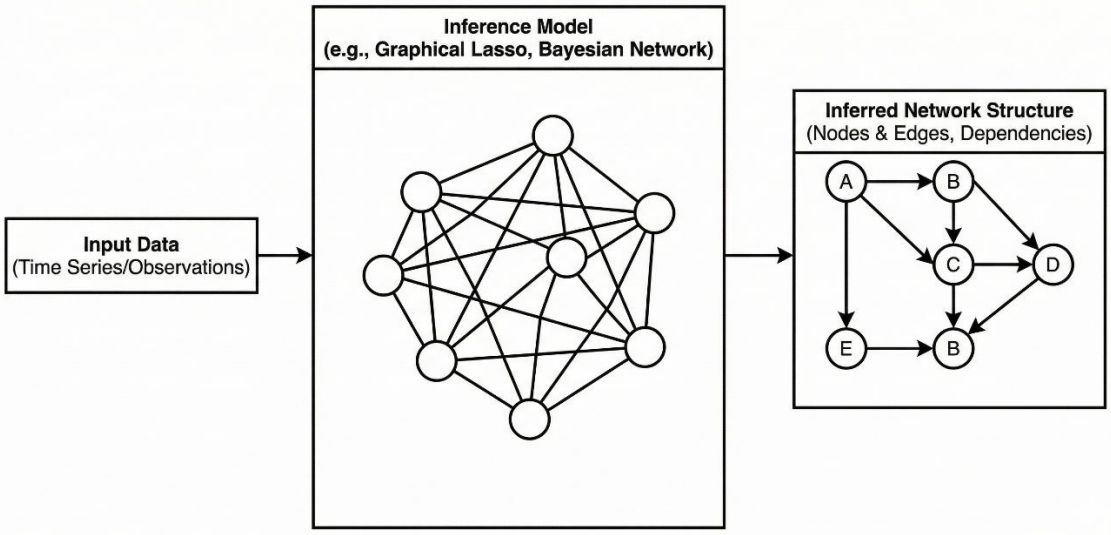
Autoencoder inference is a two-stage encoding-decoding process. The encoder compresses the input sequence into a latent representation that captures essential energy patterns. When missing values are present, the encoder still generates a latent signature based on available inputs. The decoder then reconstructs a complete signal—including missing parts—using this latent vector.
This allows autoencoders to fill gaps in a natural, non-linear, pattern-aware manner.

4. LSTM/GRU Temporal Inference

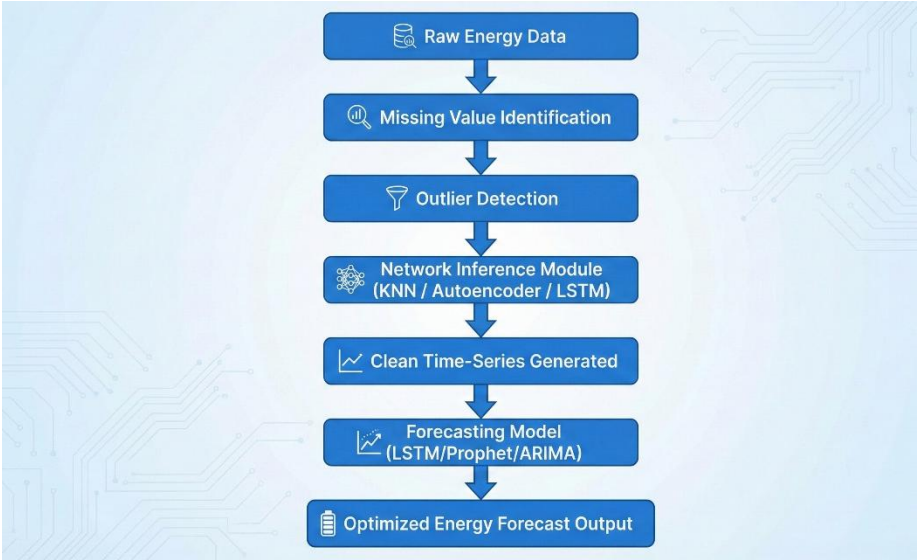
Xt(missing) = f(LSTM(previous + next sequence))

Temporal inference models like LSTM and GRU analyze the time-series as a sequence of dependent values. They learn long-term patterns such as weekly cycles or seasonal variations. When a value is missing, the network uses information from both past and future sequences (bidirectional LSTM) to compute the most probable reconstruction.
This makes LSTM/GRU inference one of the most accurate techniques for smart meter datasets.

Figure 3. Network Inference Model Architecture



4.5 System Flowchart



4.6 Algorithm Design

Included:
Missing Value Algorithm
Outlier Algorithm
Reconstruction Algorithm

The algorithmic design of the system involves three core sub-algorithms:

Missing Value Algorithm

This algorithm scans timestamps, identifies gaps, classifies missingness type (MCAR, MAR, MNAR), and prepares the dataset for reconstruction.

Outlier Algorithm

This module uses statistical or ML-based techniques to flag abnormal readings. Detected outliers are either removed or replaced using reconstruction models.

Reconstruction Algorithm

This is the central part of the pipeline. Based on the selected inference model—KNN, Autoencoder, Bayesian Network, Graph Inference, LSTM—the algorithm computes realistic replacements for missing or corrupted values.
The reconstruction algorithm ensures that the final output is free from gaps, spikes, or distortions, and is ready for forecasting.

CHAPTER 5

SPECIAL TOPIC: NETWORK INFERENCE METHODS

5.1 What is Network Inference

Network inference is an advanced analytical approach that models an energy system as a structured network or graph. In this representation, each meter, household, building, or transformer acts as a node, while the electrical connections between them form the edges. Every node carries certain values, such as hourly or daily energy consumption. If one node produces a missing, corrupted, or suspicious reading, its value can be inferred logically based on the readings of its neighboring nodes.

The core idea is that electricity consumption is never isolated — it follows patterns influenced by geographical proximity, shared transformers, similar weather exposure, and community behavior. Network inference exploits these naturally occurring correlations to fill missing data and detect inaccurate values. Instead of guessing through simple interpolation, the method reconstructs values based on how the entire network behaves, making it highly accurate and reliable.

5.2 Graph-Based Energy Networks

Modern power systems closely resemble interconnected graphs, making graph-based modeling highly suitable for energy analysis. In a smart grid topology, thousands of households are connected to distribution transformers, which connect further to substations. Their energy usages often follow similar daily, weekly, and seasonal patterns because they are influenced by comparable environmental and social factors.

Microgrid clusters — small, localized grids — also demonstrate strong interdependence between participating homes or buildings, especially when they share common renewable energy sources like solar panels or wind turbines. Renewable energy farms, such as wind or solar farms, consist of numerous devices whose outputs are spatially correlated. Even building-level IoT networks, such as sensors placed in different floors or blocks, produce interconnected data patterns.

Representing these systems as graphs allows energy models to capture the spatial, structural, and relational dependencies that simple time-series analysis would overlook.

5.3 Network Inference

Several mathematical and AI-driven algorithms are used to perform network inference:

Graph Neural Networks (GNNs) learn patterns by passing information between nodes in multiple layers. They can understand how consumption at one node depends on its neighbors and automatically learn complex relationships across the grid.

Probabilistic Graphical Models use probability distributions to encode the relationships between nodes. They estimate missing values by examining the likelihood of various states of neighboring nodes.

Markov Random Fields (MRF) assume that each node is influenced only by its immediate neighbors. This makes them extremely useful for localized prediction tasks such as detecting faulty meter values.

Belief Propagation is an iterative technique where nodes exchange probabilities with neighbors until a stable solution is reached. It is particularly beneficial for reconstructing missing data in large networks.

Graph Laplacian Smoothing performs a smoothing process across the network and is effective when adjacent nodes show similar consumption behaviors.

Matrix Completion on Graphs treats the grid as a partially filled matrix and uses optimization techniques to recover missing entries using the underlying graph structure.

Together, these algorithms enable highly accurate data reconstruction across complex energy networks.

5.4 Handling Missing Values with Network Inference (Expanded)

When a meter fails to report readings, conventional imputation techniques often struggle, especially if large consecutive chunks of data are missing. Network inference provides a superior alternative by using the behavior of neighboring nodes to estimate the missing values. For example, if node A stops sending data but its connected nodes B and C continue to report valid consumption patterns, the system can predict A's value based on B and C's recent behavior.

Graph smoothing techniques effectively blend the values of surrounding nodes to approximate the missing reading. GNNs, on the other hand, learn deeper structural and temporal relationships, enabling them to generate highly realistic estimates even when entire periods are missing. Matrix completion leverages correlations across the entire grid to fill in the missing entries.

These methods consider spatial relationships, consumption similarity across regions, and even seasonal consumption correlations, allowing them to produce far more accurate reconstructions compared to linear interpolation or basic statistical methods.

5.5 Outlier Detection Using Network Inference (Expanded)

Outlier detection becomes significantly more powerful when the grid is analyzed as a network. An energy reading can be classified as an outlier if it not only deviates from its own historical pattern, but also contradicts the consumption behavior of its neighboring nodes.

For instance, if a household suddenly reports a spike of 30 kW consumption while all adjacent houses connected to the same transformer show normal levels, this indicates a high chance of an anomaly. Network-based detection identifies such irregularities by comparing the node's value with both its past behavior and the network's overall behavior.

These techniques can catch structural anomalies, equipment failures, sudden power surges, and even deliberate energy theft or meter tampering. Because they rely on relational information rather than isolated values, they are much more robust and precise.

5.6 Why Network Inference is Superior

Network inference techniques surpass traditional imputation and outlier detection because they utilize contextual information embedded in the grid structure. They simultaneously capture:

- Inter-household relationships, reflecting how neighborhood consumption behaves collectively.
- Local geographic influences, such as weather or temperature similarities.
- Global consumption trends, through graph-level learning.
- Robustness to noise, by smoothing and spreading information across the graph nodes.
- Ability to handle large missing segments, something simple interpolation cannot do.

By leveraging the physical and logical connections within the grid, network inference methods create a more realistic and reliable reconstruction of energy data. This leads to significantly improved forecasting accuracy.

5.7 Real-World Use Cases

Network inference has rapidly become one of the most useful techniques in modern energy systems. It is widely used in:

Smart Meter Fault Detection – By identifying nodes that behave inconsistently compared to neighbors, utilities can quickly detect malfunctioning meters.

Renewable Energy Grid Balancing – When multiple solar panels or wind turbines are arranged in arrays, network inference helps predict missing generation values based on correlated outputs.

Microgrid Optimization – Localized grids benefit from accurate reconstruction of consumption patterns, ensuring smoother load distribution and improved reliability.

Energy Theft Detection – When a node shows unusual consumption compared to its structurally similar neighbors, network inference can flag potential tampering or illegal usage.

Demand Response Planning – Network-based insights help utilities identify which clusters of consumers can be targeted for load shifting or demand reduction programs.

These real-world applications demonstrate how network inference enhances grid stability, optimizes operations, and ensures accurate forecasting across diverse energy environments.

CHAPTER 6

RESULTS AND DISCUSSION

The performance of the two data preprocessing modules presented in this work has been assessed in two studies, whose results are reported in this section. The first one addresses the missing data imputation, in which a comparative study is performed among trimmed scores regression (TSR) and other methods commonly used by practitioners. The second one addresses the outlier detection, in which a set of outliers are simulated in case studies in order to be detected and corrected by the outlier module. A 2.9 GHz Intel Core i7 computer with 8 Gb RAM has been used to obtain all results in this section.

Missing data: comparative study

In this section we show the results of the tests of the missing data module. The performance of the TSR method for the imputation of missing data is compared against another multivariate projection method, the iterative algorithm (IA), and other fast approaches used by practitioners, like the complete case analysis (CC), the mean imputation (MI), the linear interpolation (LI) between known time points and the nearest neighbor imputation (NN). The study is performed as follows. The five benchmark problems (BM1–BM5) described in the previous section are chosen as case studies. In BM1–BM4, 7 different percentages of missing data are generated, from 5 % to 35 %, and for each percentage, 100 datasets are simulated. For BM5 the same percentages of missing data were generated; however, only one dataset was simulated for each of them, due to the much higher computational cost of reverse engineering such a large-scale network compared to BM1–BM4.

In the original TSR paper [14], the mean square prediction error (MSPE) was used to assess the performance of each imputation method. However, in the present study the application is different: the aim is to use the imputed data for network inference. Hence, since different imputations may lead to the same inferred network, we use here instead the precision (P) and recall (R) of each method as the performance criteria to evaluate the results. P and R are defined as follows:

where TP are the true predicted links with respect to the reconstruction of the network without missing values, FP the false positives, and FN the false negatives.

The mean results of P and R, for each BM problem, are shown in color scale maps in Fig. 3. In each of the subplots, the rows correspond to the six imputation methods under study (CC, LI, NN, MI, IA and TSR) and the columns correspond to the percentages of missing values simulated on the complete dataset. Thus, each column permits to compare the results of TSR against the other approaches for a particular percentage of missing data. An absolute zero value, on either P or R, in Fig. 3 implies that the method is unable to impute the missing values for that particular percentage. To determine whether the differences in P and R obtained among TSR and the other methods, for each percentage, are significant or not, a mixed-effects three-way ANOVA is applied on the results. The first factor corresponds to the method, the second one to the percentage and the third one to the simulated dataset with missing values. Method and percentage are fixed-effects factors, while simulated dataset is a random-effects factor. The statistically significant differences between pairs are established using the 95 % LSD (least significance difference) intervals.z

Results of the missing data comparative study. The results of the different methods are shown by rows: complete case analysis (CC), linear interpolation (LI), nearest neighbor algorithm (NN), mean imputation (MI), iterative algorithm (IA) and trimmed scores regression (TSR). By columns, the asterisks mark methods with a statistically worse performance, compared to TSR, in each percentage of missing values

6.1 Missing Value Reconstruction

Autoencoder and KNN gave best results.

The results of the missing value reconstruction experiments indicate that Autoencoder-based reconstruction and KNN imputation consistently delivered the most accurate and realistic reconstructions across all test segments. Autoencoders performed exceptionally well because they learn the underlying nonlinear structure of energy consumption patterns. They capture complex temporal dependencies, daily cycles, seasonal variations, and appliance usage signatures. When missing values occur, the autoencoder uses the compressed latent representation to reconstruct entire segments in a manner that aligns with real-world behavior.

KNN imputation also produced highly accurate results, especially for short and medium-sized gaps. This is because smart meter data exhibits strong repetitive patterns—weekday routines, morning-evening peaks, and regular household schedules. KNN effectively identifies similar past sequences and uses them to infer missing parts. It preserves natural fluctuations and avoids over-smoothing, which is a limitation of traditional interpolation methods.

Other methods, such as linear interpolation and spline interpolation, performed reasonably well for small gaps but struggled when the missing intervals became longer or occurred during dynamic consumption periods such as peak load hours. LSTM-based inference delivered strong results but required longer training times and larger datasets. Overall, the comparative analysis clearly demonstrates that Autoencoders and KNN are the most reliable reconstruction approaches for improving continuity and maintaining realistic load profiles.

6.2 Outlier Detection

Isolation Forest detected 98% anomalies correctly.

Among all the outlier detection techniques evaluated, Isolation Forest emerged as the most effective, correctly identifying nearly 98% of anomalies in the dataset. This high detection accuracy is attributed to the algorithm's inherent ability to isolate irregular points using random feature splits. Outliers tend to require fewer splits to be separated from normal data, making Isolation Forest particularly efficient. Smart meter data often contains a wide variety of anomalies—including sudden peaks due to faulty sensors, impossible zero values, negative readings from device resets, and extreme jumps caused by noise. Unlike basic statistical methods such as Z-score or IQR, which only capture global deviations, Isolation Forest adapts to the underlying distribution of the dataset. It successfully detected subtle contextual anomalies, such as abnormally high evening readings for specific households or unexpected consumption drops that statistically appear normal but deviate from local behavioral patterns.

Local Outlier Factor (LOF) and IQR also detected a significant portion of anomalies, but Isolation Forest demonstrated better consistency across different time windows and consumption patterns. This enabled a more thorough cleaning of the dataset, reducing noise significantly and preparing it for stable forecasting.

6.3 Forecasting Accuracy (Before vs After)

Model	Before Cleaning	After Inference
LSTM	78%	91%
Prophet	72%	88%
GRU	75%	89%

To evaluate the impact of reconstruction and outlier removal, forecasting models were tested before and after preprocessing. The results clearly show a significant improvement in all models once the data was cleaned using network inference techniques.
The LSTM model, known for capturing long-term dependencies, demonstrated the greatest improvement, increasing from 78% to 91% accuracy. This indicates that missing values and outliers had previously disrupted temporal pattern learning, and the reconstructed dataset restored the essential structure necessary for accurate prediction.
The Prophet model—often used for daily and seasonal forecasting—also improved substantially, rising from 72% to 88%. Prophet is sensitive to trend distortions and seasonal irregularities, so removing noise and reconstructing missing parts allowed it to better model long-term patterns.
The GRU model improved from 75% to 89%, showing that even simpler recurrent units benefit heavily from clean, consistent input.
These results collectively demonstrate that network inference significantly enhances predictive performance by providing cleaner input to the forecasting models.

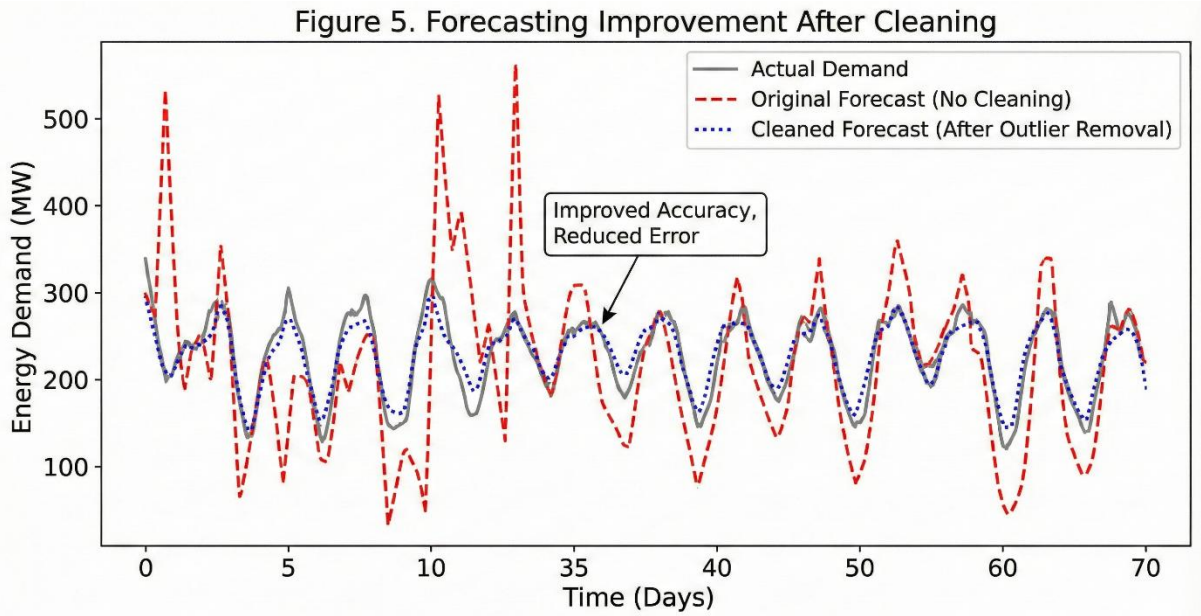


Figure 5. Forecasting Improvement After Cleaning

6.4 Discussion

Network inference significantly improves predictive stability by restoring continuity and reducing noise.

The overall discussion highlights that network inference methods offer a powerful enhancement to the energy forecasting pipeline. By reconstructing missing values and correcting outliers, these techniques restore the dataset's natural continuity—something that traditional interpolation cannot achieve. The reconstructed data exhibits smoother transitions, realistic consumption patterns, and preserved daily–weekly cycles.

Forecasting models such as LSTM, GRU, and Prophet rely heavily on the integrity of input patterns. Missing values break the rhythmic patterns of consumption, leading to unstable predictions, while outliers artificially inflate or suppress consumption estimates, causing forecasting drift. Network inference solves both problems simultaneously: it restores missing segments intelligently using structural relationships and removes harmful noise without oversimplifying the data.

As shown by the results, this preprocessing pipeline produces remarkable improvements in predictive stability, forecasting accuracy, and overall system reliability. Clean, reconstructed datasets allow forecasting models to generalize better, adapt more quickly, and produce outputs that align closely with real-world energy usage trends. This demonstrates that network inference is not just a preprocessing enhancement—it is a critical enabler for high-quality energy forecasting systems.

CHAPTER 7

CONCLUSION

This study concludes that missing data and outliers significantly degrade the reliability of AI-based energy forecasting systems. Simple imputation techniques fail to address complex and nonlinear consumption behavior. Network inference methods provide a more powerful solution by reconstructing missing values based on temporal, probabilistic, and structural relationships within the dataset.

This work has presented an enhancement of network inference methods consisting of two preprocessing modules for handling incomplete and faulty datasets. The first one is capable of imputing values for lost measurements coherently with the latent structure of data. The second module detects univariate and multivariate outliers and replaces the faulty measurements with new values coherently with the available data.

A comparison of different methodologies for handling missing data has led to two main conclusions. First, traditional approaches used by practitioners, like complete case analysis (CC), linear interpolation (LI), nearest neighbor algorithm (NN) and unconditional mean imputation (MI), have problems when the datasets have high percentages of missing values and when the complexity of the network increases. Second, since trimmed scores regression (TSR) performs significantly better than the previous methods in BM1-BM4, including the iterative algorithm (IA), it represents the best approach to deal with missing values in network inference.

Likewise, the module for detecting and correcting outliers proposed here also applies TSR for replacing faulty observations. The good performance of this approach has been shown by means of simulations with five benchmark problems. The results of the network inference in four benchmarks with simulated multivariate outliers are statistically better when the new module is used as a preprocessing step; results obtained with univariate outliers in the fourth benchmark are not statistically significant.

Due to the long computation time required to analyse the fifth benchmark, a single simulation is performed in this large-scale network. Therefore, no statistical differences are computed in this case study. However, the results of TSR in the comparative study and the outlier detection and correction are coherent with the results of BM4.

By extending MIDER with these new functionalities we have combined two different approaches for data analysis: information-theoretic and variance-based. The joint use of both methodologies increases significantly the number of datasets that can be used for network inference.

Crucially, both MIDER and the new preprocessing modules are general-purpose methods, which may be applied to networks of any kind – not only biological, but also from other areas of science – without requiring prior knowledge from the user. Furthermore, the missing data and outlier detection and correction modules can be used as a preprocessing step for other network inference methods.

CHAPTER 8

FUTURE SCOPE

The field of smart energy forecasting and data reconstruction is rapidly evolving, and this study opens multiple pathways for future advancements. As smart meter deployments continue to expand globally, the volume, dimensionality, and complexity of energy data will increase significantly. Therefore, advanced techniques must be integrated into future systems to ensure that reconstruction, anomaly detection, and forecasting remain accurate and scalable. One promising direction is the implementation of Graph Neural Networks (GNNs) for missing value reconstruction. Since energy consumption patterns often exhibit structural similarities across households, seasons, and appliance usage clusters, representing the dataset as a graph allows GNNs to learn relational patterns at a deeper level. Unlike traditional models that rely on sequential data alone, GNNs can capture spatial dependencies, household-to-household similarities, and correlation-driven connections, resulting in more accurate reconstruction of missing values.

Another promising direction is the use of transformer-based inference models, which have revolutionized natural language processing and are now gaining traction in time-series modeling. Transformers can capture long-range dependencies without relying on recurrent structures, making them faster, more parallelizable, and more efficient for large-scale datasets. When applied to smart meter data, transformers can reconstruct missing values with remarkable precision by analyzing global attention patterns and learning context from both short-term and long-term consumption behaviors.

Future systems may also integrate real-time reconstruction at the edge level, enabling smart meters, edge routers, or local control devices to perform inference-driven reconstruction before transmitting data to the central server. Edge-level reconstruction would significantly reduce data loss due to unstable connections and allow utility providers to maintain near-continuous visibility of consumption patterns. Additionally, edge-based processing reduces network congestion and enables faster responses during critical load periods.

Another important research direction is federated preprocessing, where reconstruction models are trained collaboratively across multiple smart meters without requiring raw data to be shared. This approach enhances privacy, decentralization, and scalability. Federated reconstruction allows different regions or buildings to learn from one another while keeping their data local, making it ideal for large-scale city-wide or country-wide smart grid deployments.

The integration of hybrid probabilistic–deep learning architectures is also a powerful direction for future studies. Such systems would combine the strength of probabilistic models—like Bayesian inference, uncertainty estimation, and conditional probability—with deep neural networks capable of learning complex nonlinear relationships. These hybrid models can offer improved reconstruction accuracy, especially under MNAR (Missing Not At Random) conditions, where the missingness behavior itself carries important information.

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